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Course Code/Course Name: DSA06 /Data Handling and Visualization

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LIST OF EXPERIMENTS

13. Dataset: Geographic Data

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

Install packages (run once)

```
install.packages("leaflet")
```

Load library

```
library(leaflet)
```

Create dataset

```
city <- c("City A", "City B", "City C")
```

```
population <- c(500000, 700000, 600000)
```

```
temperature <- c(75, 68, 80)
```

```
elevation <- c(1000, 800, 1200)
```

Sample latitude and longitude

```
lat <- c(13.0827, 12.9716, 11.0168)
```

```
lon <- c(80.2707, 77.5946, 76.9558)
```

```
data <- data.frame(city, population, temperature, elevation, lat, lon)
```

```
# Create map
```

```
leaflet(data) %>%
```

```
  addTiles() %>%
```

```
  addCircleMarkers(
```

```
    lng = ~lon,
```

```
    lat = ~lat,
```

```
    radius = ~population / 100000,
```

```
    color = "blue",
```

```
    fillOpacity = 0.7,
```

```
    popup = ~paste(
```

```
      "<b>", city, "</b><br>",
```

```
      "Population:", population, "<br>",
```

```
      "Avg. Temperature:", temperature, "°F<br>",
```

```
      "Elevation:", elevation, " m"
```

```
    )
```

```
  ) %>%
```

```
  addLegend(
```

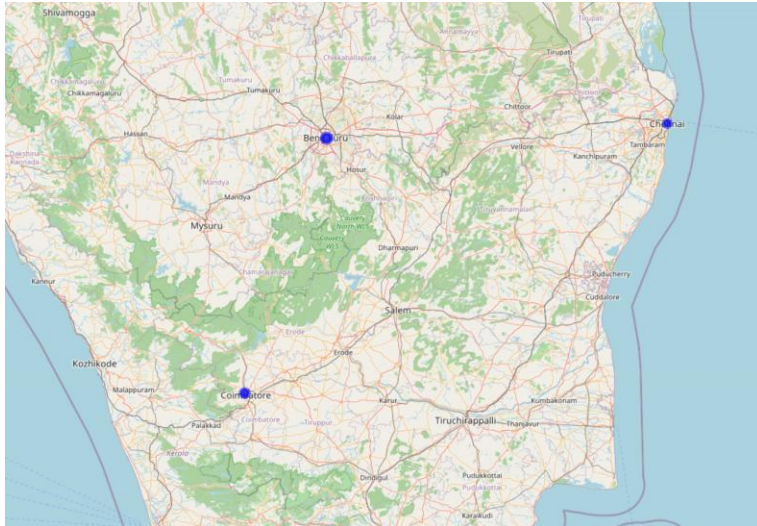
```
    position = "bottomright",
```

```
    colors = "blue",
```

```
    labels = "City Locations",
```

```
    title = "Cities"
```

```
  )
```



2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

```
# Create dataset
```

```
city <- c("City A", "City B", "City C")
```

```
population <- c(500000, 700000, 600000)
```

```
temperature <- c(75, 68, 80)
```

```
elevation <- c(1000, 800, 1200)
```

```
data <- data.frame(city, population, temperature, elevation)
```

```
# Scatter Plot
```

```
plot(
```

```
  data$temperature,
```

```
  data$population,
```

```
  main = "Average Temperature vs Population",
```

```
  xlab = "Average Temperature (°F)",
```

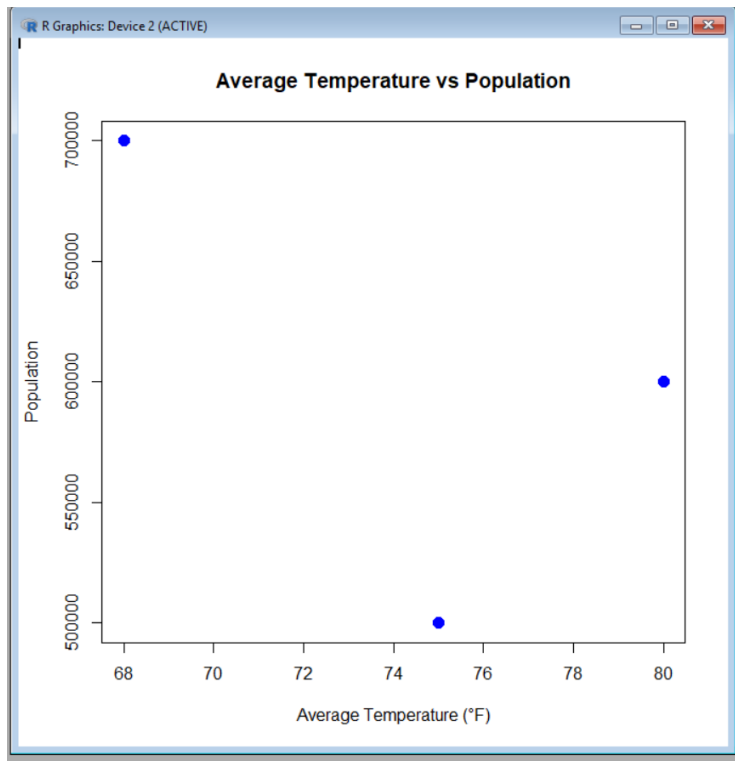
```
  ylab = "Population",
```

```
  pch = 19,
```

```
col = "blue",  
cex = 1.5  
)  
  
# Add city labels  
text(  
  data$temperature,  
  data$population,  
  labels = data$city,  
  pos = 4,  
  cex = 0.8  
)
```

Insights

-  **City B** has the **highest population (700,000)** but the **lowest average temperature (68°F)**.
-  **City C** has the **highest average temperature (80°F)** with a **moderate population (600,000)**.
-  **City A** falls between the other two in both temperature and population.
-  With only **three data points**, there is **no clear positive or negative relationship** between average temperature and population. A larger dataset would be needed to identify any meaningful trend.



3. Build a table to display the geographic data for each city. Label the table elements.

```
# Create dataset
```

```
city <- c("City A", "City B", "City C")
```

```
population <- c(500000, 700000, 600000)
```

```
temperature <- c(75, 68, 80)
```

```
elevation <- c(1000, 800, 1200)
```

```
# Create data frame
```

```
geographic_data <- data.frame(
```

```
  City = city,
```

```
  Population = population,
```

```
  Avg_Temperature = temperature,
```

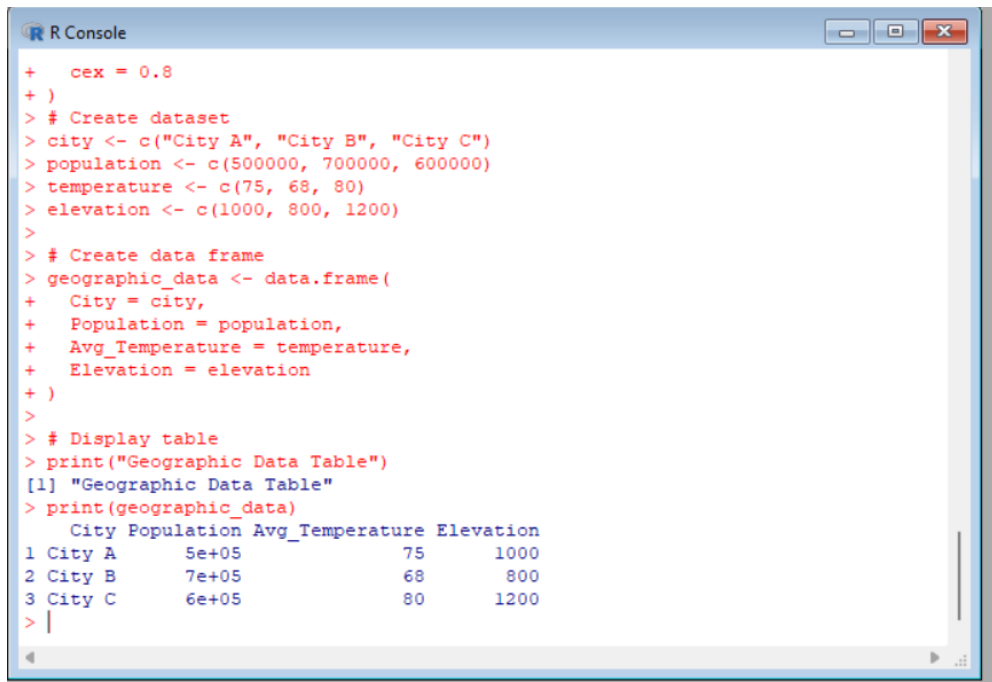
```
  Elevation = elevation
```

```
)
```

Display table

```
print("Geographic Data Table")
```

```
print(geographic_data)
```



```
+ cex = 0.8
+ )
> # Create dataset
> city <- c("City A", "City B", "City C")
> population <- c(500000, 700000, 600000)
> temperature <- c(75, 68, 80)
> elevation <- c(1000, 800, 1200)
>
> # Create data frame
> geographic_data <- data.frame(
+   City = city,
+   Population = population,
+   Avg_Temperature = temperature,
+   Elevation = elevation
+ )
>
> # Display table
> print("Geographic Data Table")
[1] "Geographic Data Table"
> print(geographic_data)
  City Population Avg_Temperature Elevation
1 City A      5e+05             75      1000
2 City B      7e+05             68       800
3 City C      6e+05             80      1200
> |
```

4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive

exploration of geographic data.

Tableau Dashboard: Geographic Data

Dashboard Title: Geographic Data Analysis Dashboard

Steps to Create the Dashboard

1. Import the Dataset

- Open Tableau.
- Connect to the Geographic Data dataset (Excel/CSV).

2. Create Sheet 1 – Map Chart

- Drag **City** to the canvas.

- Assign the Geographic Role (or use Latitude and Longitude if available).
- Drag **Population** to **Size**.
- Drag **Avg. Temperature** and **Elevation** to **Tooltip**.
- Rename the sheet **Map Chart**.

3. Create Sheet 2 – Scatter Plot

- Drag **Avg. Temperature** to **Columns**.
- Drag **Population** to **Rows**.
- Drag **City** to **Label**.
- Choose **Circle** as the Marks type.
- Rename the sheet **Scatter Plot**.

4. Create Sheet 3 – Table

- Drag **City**, **Population**, **Avg. Temperature**, and **Elevation** to the **Text** shelf.
- Rename the sheet **Geographic Data Table**.

5. Create Dashboard

- Click **New Dashboard**.
- Set the dashboard size to **Automatic**.
- Drag the three sheets onto the dashboard:
 -  **Map Chart** (Top Left)
 -  **Scatter Plot** (Top Right)
 -  **Geographic Data Table** (Bottom)

6. Add Interactivity

- Go to **Dashboard** → **Actions** → **Add Action** → **Filter**.
- Source Sheets: **Map Chart** and **Scatter Plot**.
- Target Sheets: **Scatter Plot** and **Geographic Data Table**.
- Run Action: **Select**.

7. Format Dashboard

- Add the title "**Geographic Data Analysis Dashboard**".
- Show legends for **Population**.

- Display tooltips with:
 - City
 - Population
 - Average Temperature
 - Elevation

Final Dashboard Layout

Geographic Data Analysis Dashboard

Map Chart

Scatter Plot

Geographic Data Table

Dashboard Features

-  Interactive **Map Chart** displaying city locations.
-  **Scatter Plot** showing the relationship between **Average Temperature** and **Population**.
-  **Data Table** listing all city information.
-  Selecting a city in the map or scatter plot filters and highlights the corresponding record in the table for interactive exploration.

14. Dataset: Survey Results

1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

```
# Create dataset
```

```
respondent <- c(1, 2, 3)
```

```
Q1 <- c("A", "B", "C")
```



```
Q2 <- c("B", "A", "A")
```

```
Q3 <- c("C", "D", "B")
```

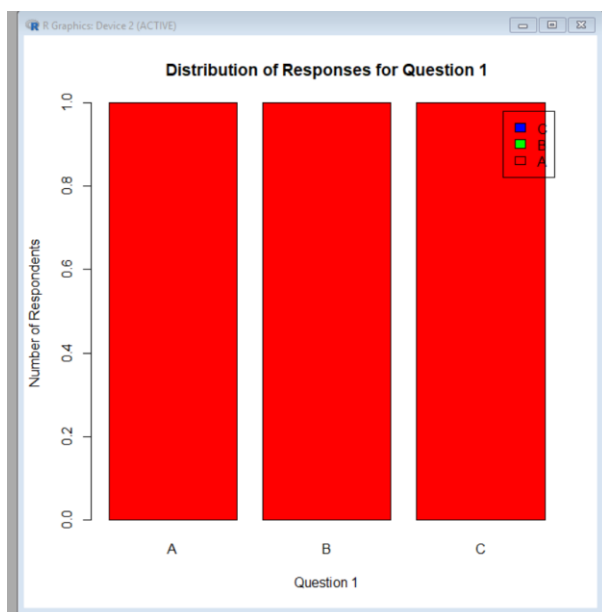
```
survey <- data.frame(respondent, Q1, Q2, Q3)
```

```
# Count responses for Question 1
```

```
q1_count <- table(survey$Q1)
```

```
# Create stacked bar chart
```

```
barplot(  
  rbind(q1_count),  
  main = "Distribution of Responses for Question 1",  
  xlab = "Question 1",  
  ylab = "Number of Respondents",  
  col = c("red", "green", "blue"),  
  legend.text = names(q1_count)  
)
```



2. Generate a radar chart to visualize the overall pattern of responses across all three questions.

```
# Install package (run once)
```

```
install.packages("fmsb")
```

```
# Load library
```

```
library(fmsb)
```

```
# Count responses for each question
```

```
A <- c(1, 1, 1) # Q1, Q2, Q3
```

```
B <- c(1, 1, 1)
```

```
C <- c(1, 0, 1)
```

```
D <- c(0, 1, 1)
```

```
# Create data frame
```

```
data <- data.frame(
```

```
  Q1 = c(3, 0, A[1], B[1], C[1], D[1]),
```

```
  Q2 = c(3, 0, A[2], B[2], C[2], D[2]),
```

```
  Q3 = c(3, 0, A[3], B[3], C[3], D[3])
```

```
)
```

```
rownames(data) <- c("Max", "Min", "A", "B", "C", "D")
```

```
# Create radar chart
```

```
radarchart(
```

```
  data,
```

```
  axistype = 1,
```

```
  pcol = c("red", "blue", "green", "purple"),
```

```

plwd = 2,
plty = 1,
title = "Overall Pattern of Survey Responses"
)

```

```

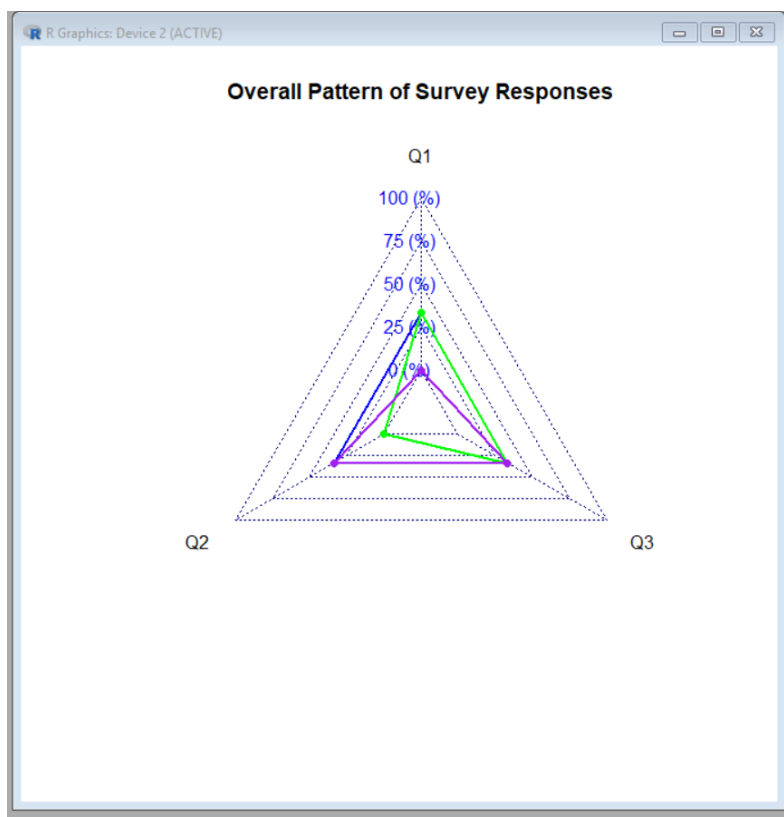
# Add legend

```

```

legend(
  "topright",
  legend = c("A", "B", "C", "D"),
  col = c("red", "blue", "green", "purple"),
  lty = 1,
  lwd = 2
)

```



3. Build a table to show the survey response data for each respondent. Label the table elements.

```
# Create dataset
```

```
Respondent <- c(1, 2, 3)
```

```
Question1 <- c("A", "B", "C")
```

```
Question2 <- c("B", "A", "A")
```

```
Question3 <- c("C", "D", "B")
```

```
# Create data frame
```

```
survey_data <- data.frame(
```

```
  Respondent,
```

```
  Question1,
```

```
  Question2,
```

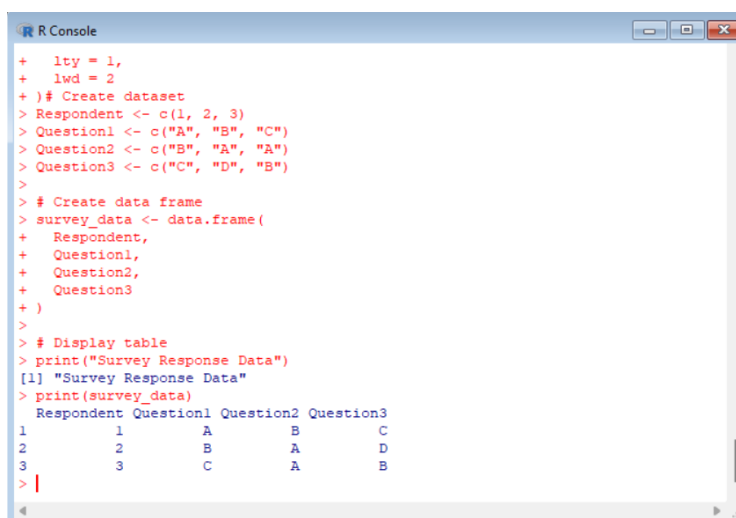
```
  Question3
```

```
)
```

```
# Display table
```

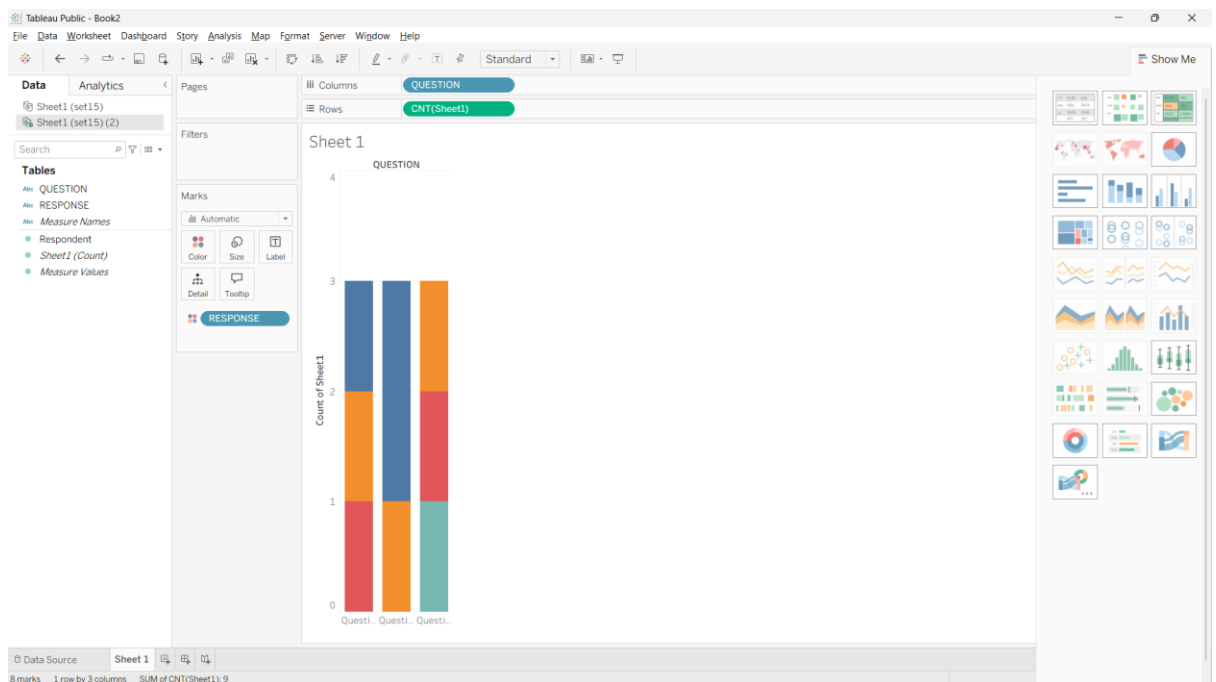
```
print("Survey Response Data")
```

```
print(survey_data)
```



```
R Console
+ lty = 1,
+ lwd = 2
+ )# Create dataset
> Respondent <- c(1, 2, 3)
> Question1 <- c("A", "B", "C")
> Question2 <- c("B", "A", "A")
> Question3 <- c("C", "D", "B")
>
> # Create data frame
> survey_data <- data.frame(
+   Respondent,
+   Question1,
+   Question2,
+   Question3
+ )
>
> # Display table
> print("Survey Response Data")
[1] "Survey Response Data"
> print(survey_data)
  Respondent Question1 Question2 Question3
1          1         A         B         C
2          2         B         A         D
3          3         C         A         B
```

4. Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.



5.

Set 15: Product Sales Analysis

Dataset: Monthly Product Sales

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

Install ggplot2 (Run only once)

```
install.packages("ggplot2")
```

Load library

```
library(ggplot2)
```

Create dataset

```
sales <- data.frame(
```

```
  Product = c("Product A", "Product B", "Product C"),
```

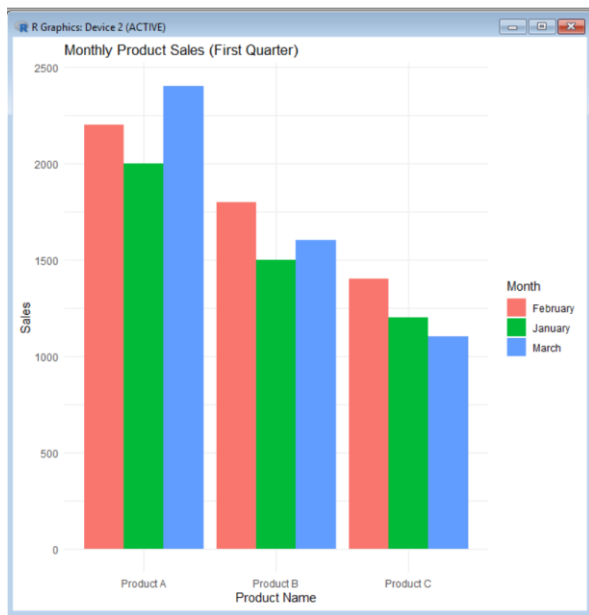
```

January = c(2000, 1500, 1200),
February = c(2200, 1800, 1400),
March = c(2400, 1600, 1100)
)

# Convert data from wide to long format
sales_long <- data.frame(
  Product = rep(sales$Product, 3),
  Month = rep(c("January", "February", "March"), each = 3),
  Sales = c(sales$January, sales$February, sales$March)
)

# Create grouped bar chart
ggplot(sales_long, aes(x = Product, y = Sales, fill = Month)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Monthly Product Sales (First Quarter)",
    x = "Product Name",
    y = "Sales",
    fill = "Month"
  ) +
  theme_minimal()

```



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

Install packages (Run only once)

```
install.packages("ggplot2")
```

```
install.packages("reshape2")
```

Load libraries

```
library(ggplot2)
```

```
library(reshape2)
```

Create dataset

```
sales <- data.frame(
```

```
  Month = c("January", "February", "March"),
```

```
  Product_A = c(2000, 2200, 2400),
```

```
  Product_B = c(1500, 1800, 1600),
```

```
  Product_C = c(1200, 1400, 1100)
```

```
)
```

```
# Convert data from wide to long format
```

```
sales_long <- melt(sales, id.vars = "Month",  
  variable.name = "Product",  
  value.name = "Sales")
```

```
# Create stacked area chart
```

```
ggplot(sales_long, aes(x = Month, y = Sales, fill = Product, group = Product)) +  
  geom_area(alpha = 0.8) +  
  labs(  
    title = "Overall Product Sales Trend (First Quarter)",  
    x = "Month",  
    y = "Sales",  
    fill = "Product"  
  ) +  
  theme_minimal()
```



3. Build a table to show the monthly sales data for each product. Label the table elements.

Install package (Run only once)

```
install.packages("gridExtra")
```

Load library

```
library(gridExtra)
```

Create dataset

```
sales <- data.frame(
```

```
  Product_ID = c(1, 2, 3),
```

```
  Product_Name = c("Product A", "Product B", "Product C"),
```

```
  January_Sales = c(2000, 1500, 1200),
```

```
  February_Sales = c(2200, 1800, 1400),
```

```
  March_Sales = c(2400, 1600, 1100)
```

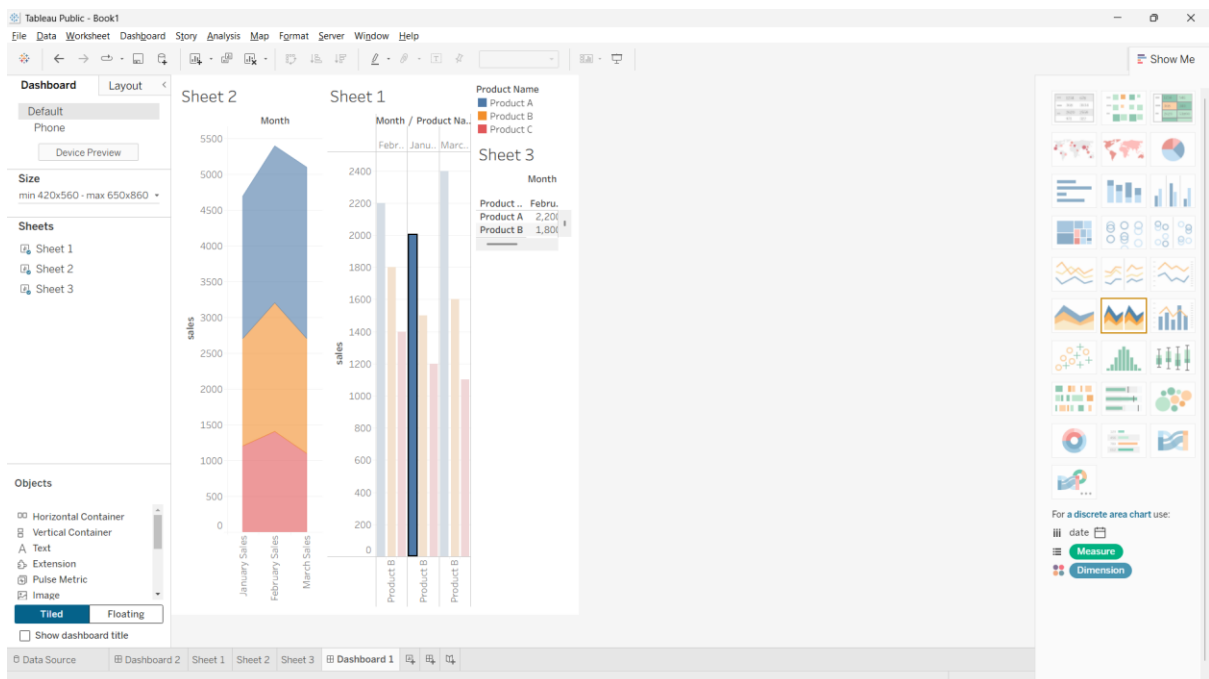
```
)
```

Display the table

```
grid.table(sales)
```

R Graphics: Device 2 (ACTIVE)					
	Product_ID	Product_Name	January_Sales	February_Sales	March_Sales
1	1	Product A	2000	2200	2400
2	2	Product B	1500	1800	1600
3	3	Product C	1200	1400	1100

4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.



Set 16: Customer Demographics Analysis

Dataset: Customer Demographics

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

```
# Customer Demographics Data  
customer <- c("Customer 1", "Customer 2", "Customer 3")  
age <- c(28, 35, 42)
```

```
# Create Bar Chart  
barplot(age,  
  names.arg = customer,  
  col = c("pink", "lightblue", "lightgreen"),  
  main = "Customer Age Distribution",  
  xlab = "Customer ID",  
  ylab = "Age (Years)")
```

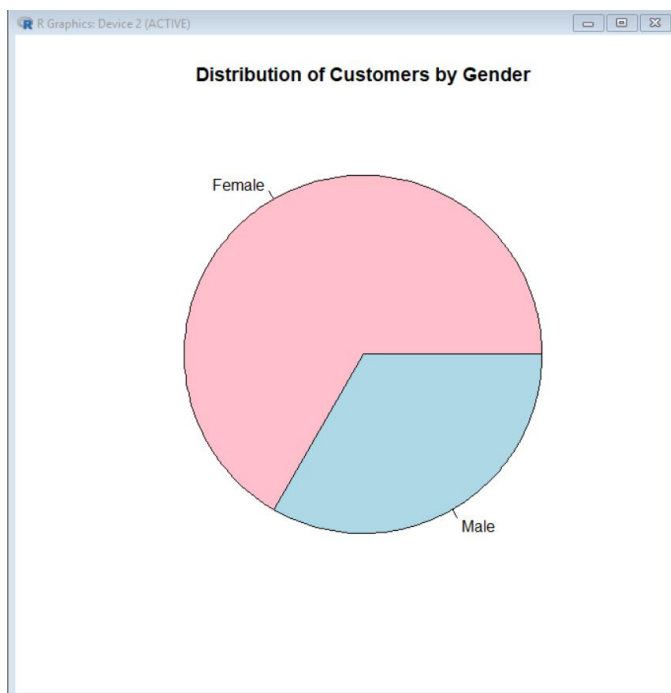


2. Generate a pie chart to represent the distribution of customers by gender.

```
# Customer Gender Data  
gender <- c("Female", "Male")  
count <- c(2, 1)
```

```
# Create Pie Chart
```

```
pie(count,  
     labels = gender,  
     col = c("pink", "lightblue"),  
     main = "Distribution of Customers by Gender")
```



3. Build a table to show the demographic information for each customer. Label the table elements.

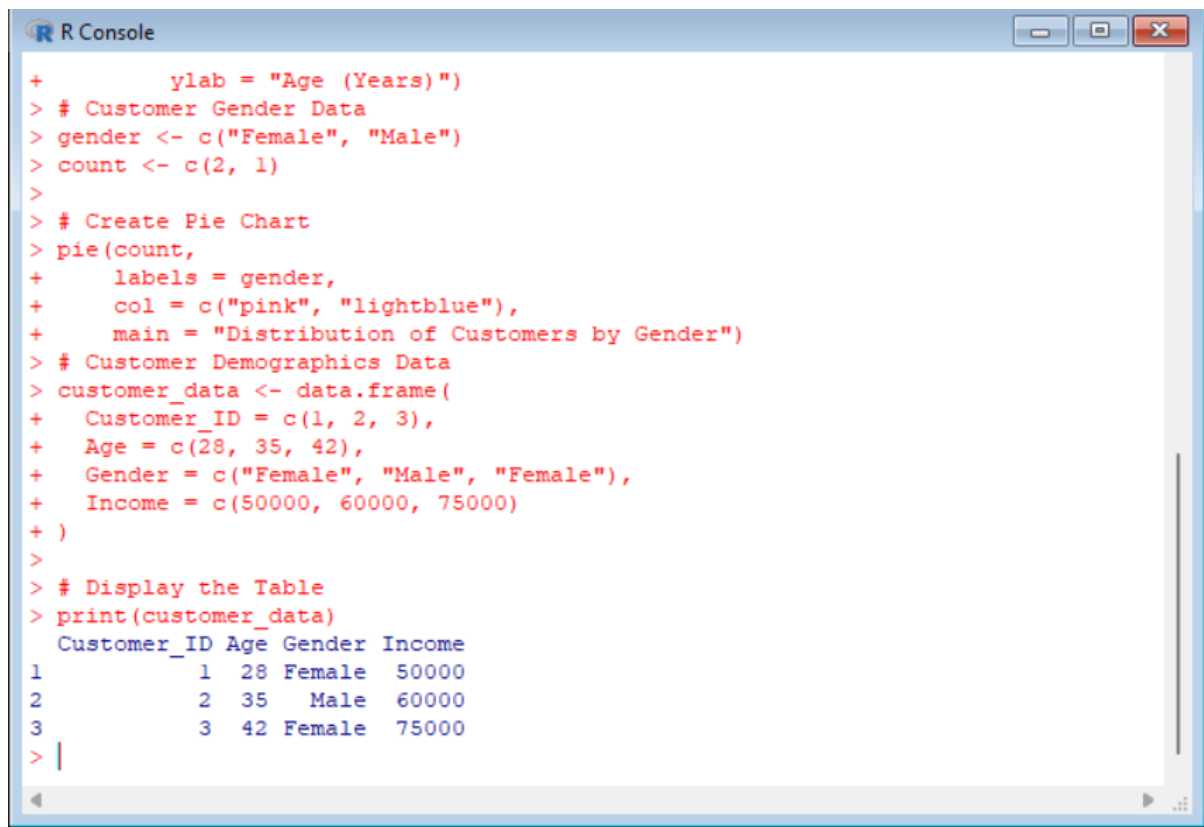
```
# Customer Demographics Data
```

```
customer_data <- data.frame(  
  Customer_ID = c(1, 2, 3),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),  
  Income = c(50000, 60000, 75000)
```

)

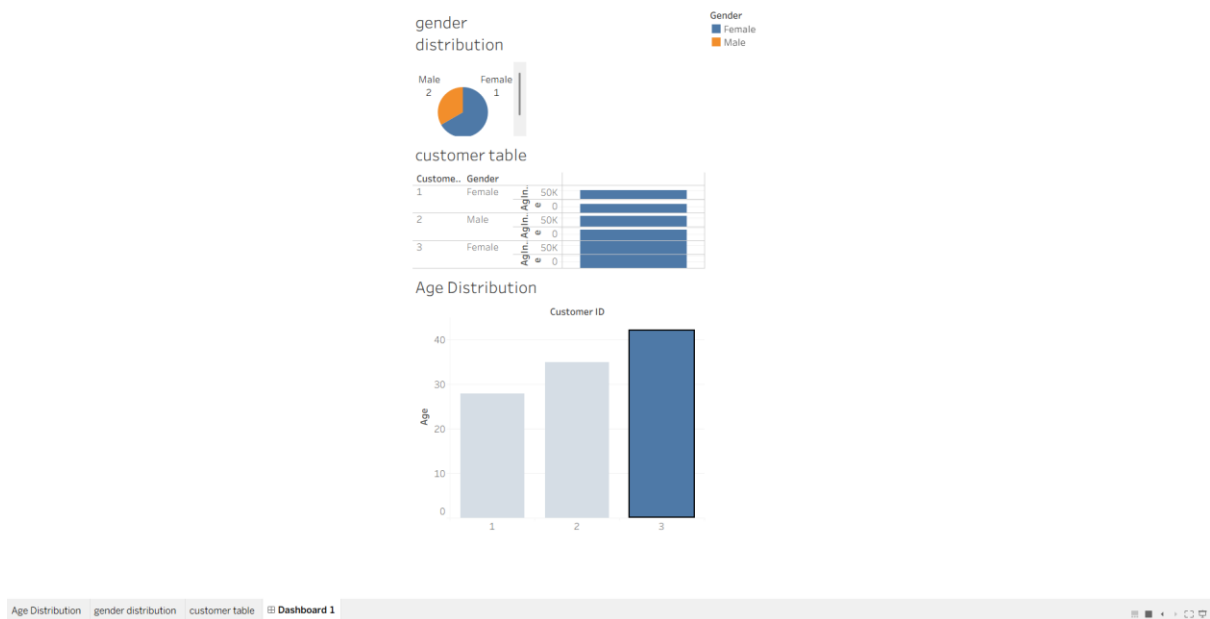
Display the Table

print(customer_data)

The image shows a screenshot of an R Console window. The window has a title bar that says "R Console" and standard Windows window controls (minimize, maximize, close). The console contains several lines of R code in red text. The code defines a variable 'ylab', creates a vector 'gender' with values "Female" and "Male", and a vector 'count' with values 2 and 1. It then uses the 'pie' function to create a pie chart with labels from 'gender' and colors "pink" and "lightblue", with a main title "Distribution of Customers by Gender". Next, it creates a data frame 'customer_data' with columns 'Customer_ID', 'Age', 'Gender', and 'Income'. The data frame contains three rows of data. Finally, it uses 'print(customer_data)' to display the data frame. The output of the print command is shown in blue text, displaying the data frame structure and its contents. The console ends with a red prompt character '>' and a vertical cursor bar.

```
+       ylab = "Age (Years)")
> # Customer Gender Data
> gender <- c("Female", "Male")
> count <- c(2, 1)
>
> # Create Pie Chart
> pie(count,
+     labels = gender,
+     col = c("pink", "lightblue"),
+     main = "Distribution of Customers by Gender")
> # Customer Demographics Data
> customer_data <- data.frame(
+   Customer_ID = c(1, 2, 3),
+   Age = c(28, 35, 42),
+   Gender = c("Female", "Male", "Female"),
+   Income = c(50000, 60000, 75000)
+ )
>
> # Display the Table
> print(customer_data)
  Customer_ID Age Gender Income
1           1  28 Female 50000
2           2  35  Male 60000
3           3  42 Female 75000
> |
```

4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.



Set 17: Employee Performance Analysis

Dataset: Employee Performance

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

Employee Performance Data

```
years <- c(3, 5, 7)
```

```
performance <- c(92, 85, 78)
```

Create Line Chart

```
plot(years, performance,
```

```
  type = "o",
```

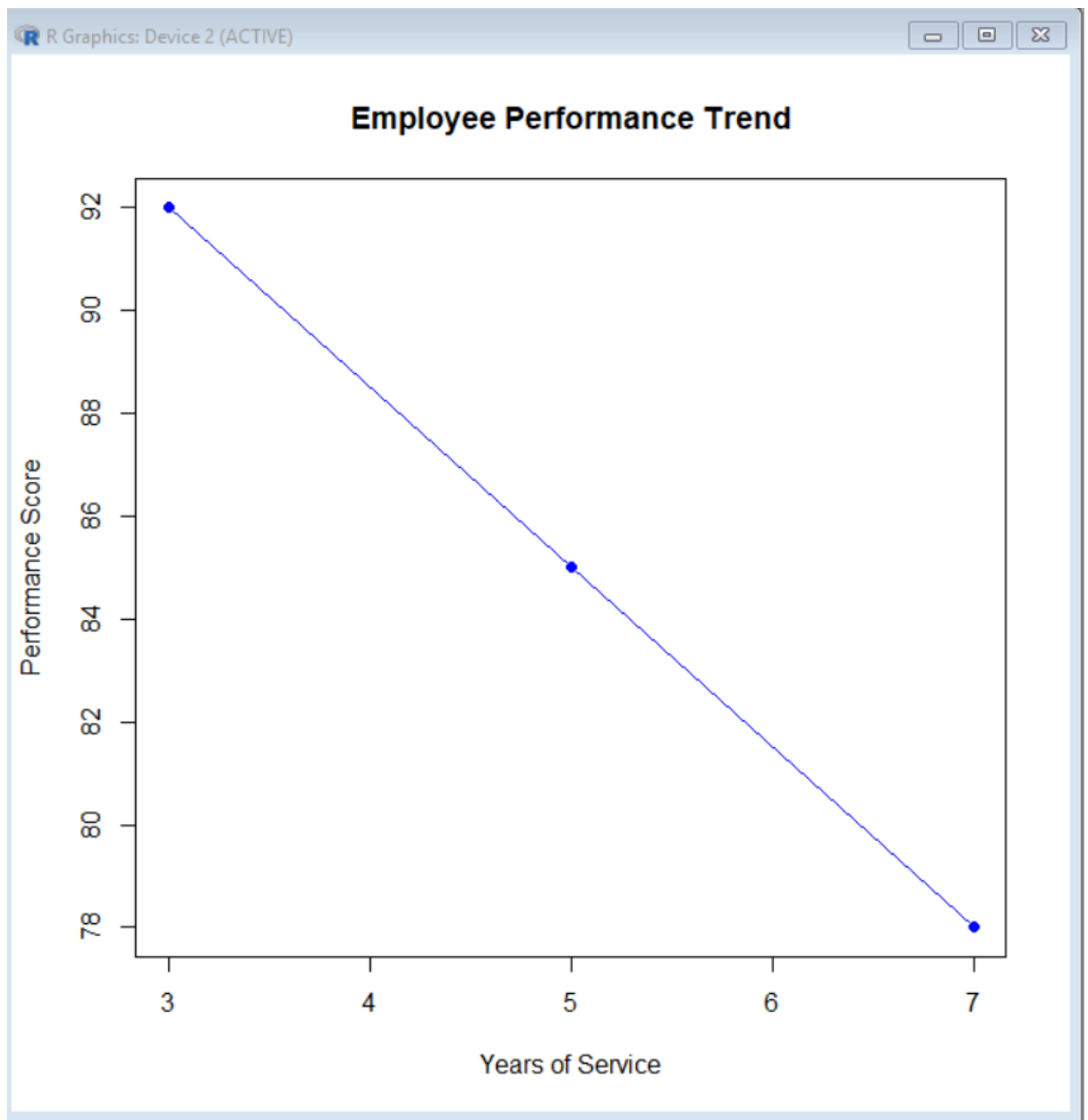
```
  col = "blue",
```

```
  pch = 16,
```

```
  xlab = "Years of Service",
```

```
  ylab = "Performance Score",
```

```
  main = "Employee Performance Trend")
```



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
# Employee Department Data
```

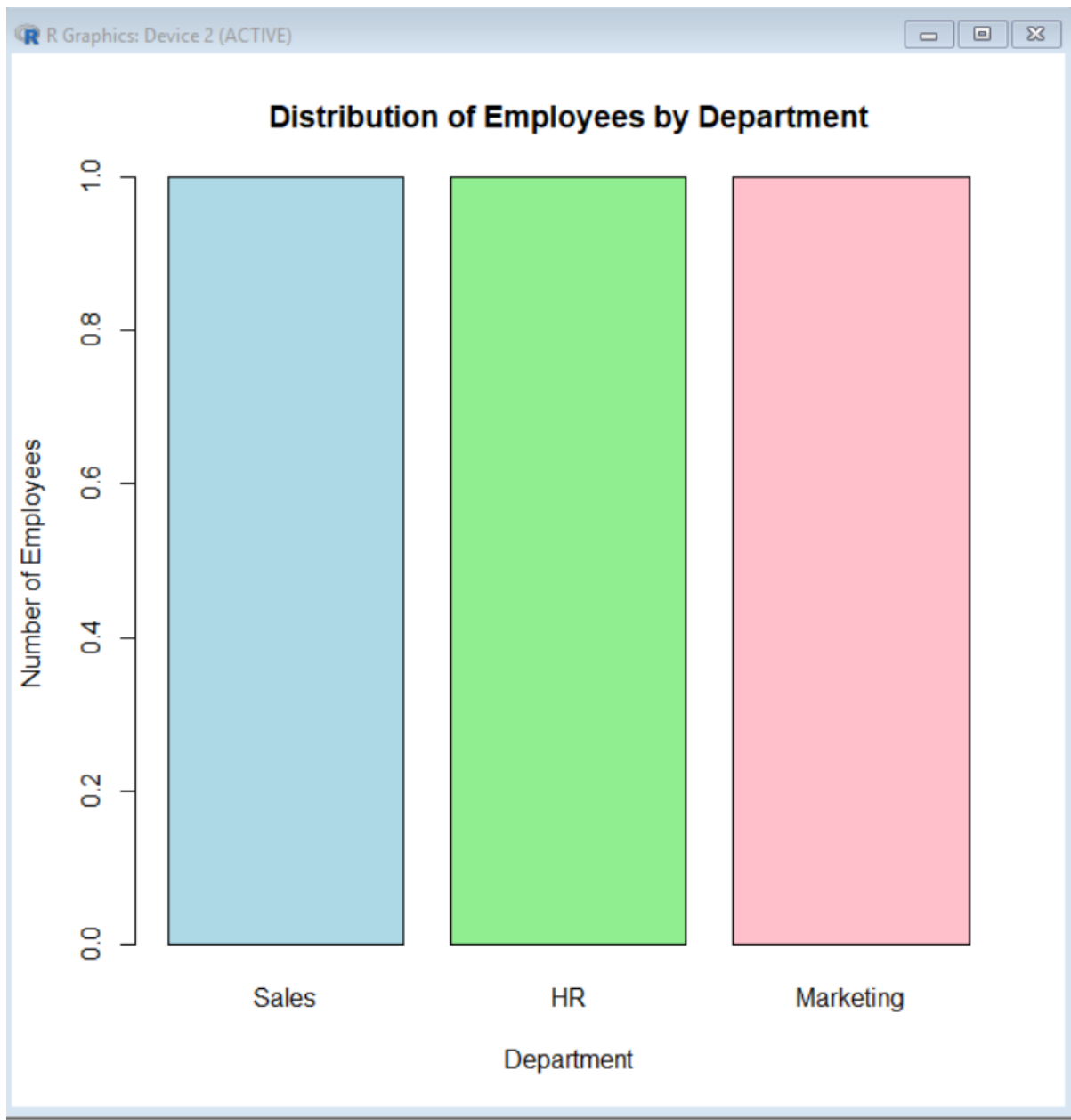
```
department <- c("Sales", "HR", "Marketing")
```

```
count <- c(1, 1, 1)
```

```
# Create Bar Chart
```

```
barplot(count,
```

```
names.arg = department,  
col = c("lightblue", "lightgreen", "pink"),  
main = "Distribution of Employees by Department",  
xlab = "Department",  
ylab = "Number of Employees")
```



3. Build a table to display the performance data for each employee. Label the table elements.


```
# Employee Performance Data

employee_data <- data.frame(

  Employee_ID = c(1, 2, 3),

  Department = c("Sales", "HR", "Marketing"),

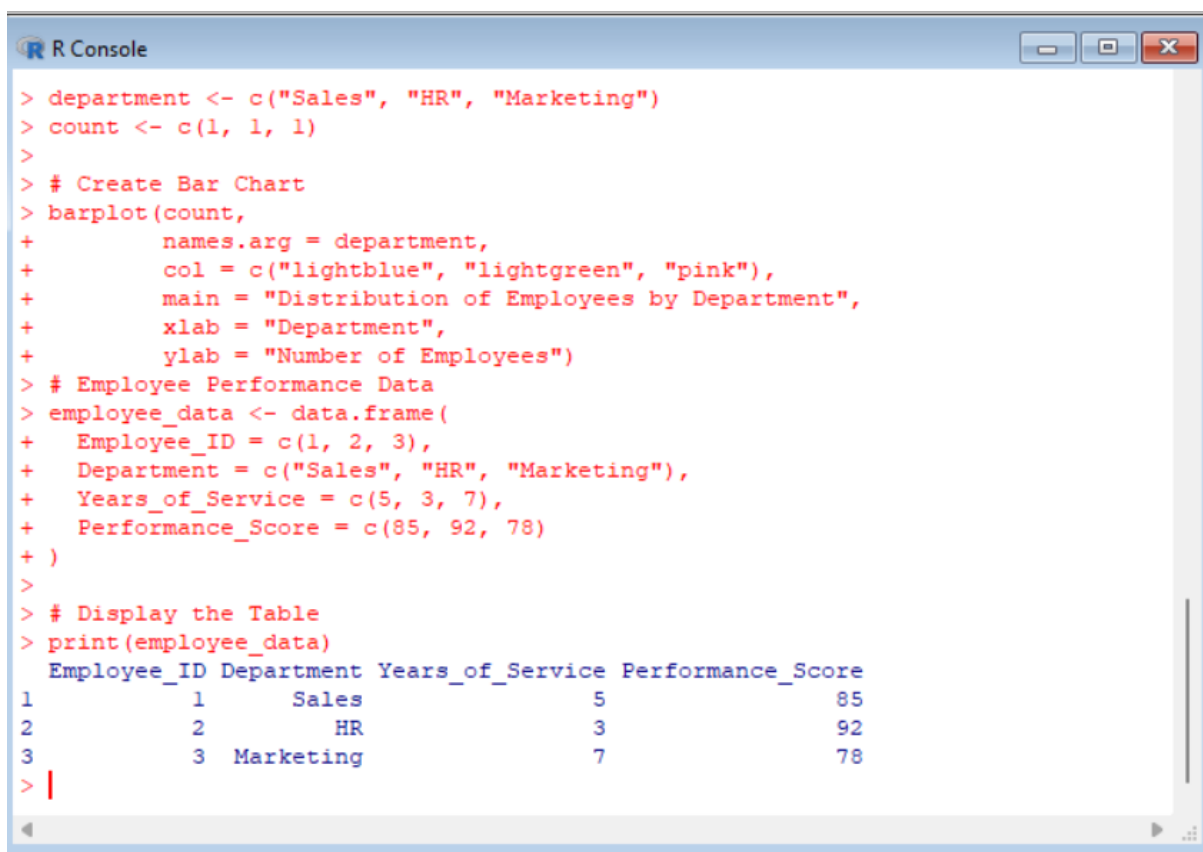
  Years_of_Service = c(5, 3, 7),

  Performance_Score = c(85, 92, 78)

)
```

```
# Display the Table

print(employee_data)
```



The screenshot shows an R Console window with the following code and output:

```
> department <- c("Sales", "HR", "Marketing")
> count <- c(1, 1, 1)
>
> # Create Bar Chart
> barplot(count,
+         names.arg = department,
+         col = c("lightblue", "lightgreen", "pink"),
+         main = "Distribution of Employees by Department",
+         xlab = "Department",
+         ylab = "Number of Employees")
> # Employee Performance Data
> employee_data <- data.frame(
+   Employee_ID = c(1, 2, 3),
+   Department = c("Sales", "HR", "Marketing"),
+   Years_of_Service = c(5, 3, 7),
+   Performance_Score = c(85, 92, 78)
+ )
>
> # Display the Table
> print(employee_data)
```

	Employee_ID	Department	Years_of_Service	Performance_Score
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

5. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

